1.1 Transcript

Hi everyone. In this module we're going to focus on prompt engineering.

And the goal here is to this is a no code module.

So there's no coding at all. The goal here is to see what we can do with a large language model.

We're going to be using ChatGPT the free version mostly except for a little bit at the end where I will be using the ChatGPT plus version.

And the goal here is it doesn't really matter whether it's ChatGPT or something else, because almost all of them have a similar structure in the way you interact with them.

So if you have ChatGPT open an account with it, it's free.

You can follow along as we work on this module.

So what we are doing is we're using large language models and we have already looked at them.

We know how they are built, we know how they're developed. So we have a good idea of what they are so we can we know that they are an AI system.

They built with huge amounts, terabytes of text data.

And the idea there is that you take in, especially in OpenAI's GPT models, you take every possible thing that is available anywhere and throw it into ChatGPT, into the sorry, into the, um, into the model, and it learns how to, uh, it's essentially it learns the totality of human knowledge.

And that's the idea here, is learning the totality of human knowledge in a structured format in this network through, uh, weights.

And, um, you know, everything is probabilistic.

Finally, the weights and everything, when you give it a new input, the weights fire and you get some output which says the next word is this, the next word is this, then the next word is this, and so on and so forth.

Until you get your like an essay or an email or something like that.

Uh, they're built using transformer models, technically transformer models, plus other models as well.

So but the base is often just transformer model. Uh, they build response sequences.

And the idea is like we talked about that the basic principle here is the next word prediction.

If you think about writing an email, you're going to say, all right. Dear so-and so,

Right. So, Dear Jack, say, I need you each.

And, uh, well, why haven't you responded to my email?

So each word that you're writing, you're predicting sort of what the next word is going to be, "why" and then "haven't" then "you" and then "responded."

So similarly, what these AI LLMs are doing is they're sort of starting with the first word and then predicting the next one, and then the next one and the next one.

And they have all this knowledge that is being collected over here that helps them, you know, figure out what to say for any given context.

They are foundational models in the sense that they're general purpose.

Okay. So what does that mean? That means that you they're really collecting knowledge for whatever it's given.

So everything and if you have a specific purpose then they are not really tuned to that.

They can answer questions about almost anything.

But then you have to tailor them or fine tune them for your specific purpose because they are going to give you very generic answers, right?

We will see examples of how that works too. And they use a chat format, which is sort of really nice because you can ask a question, you get an answer, you can say, hey, I don't like this answer, or can you elaborate on it?

Or can you summarize it? Or can you be briefer, or can you focus on the important thing and it it'll do that.

Right. So it's a, it's a chat. Um, it's a query response kind of system.

But it's like a chat. Right. It's really like a it's a chat bot essentially.

Large language models are multimodal.

You can converse with them through text, through sound, through images, through video, in the sense that you can give it any one of these things, and you can get a response that is in sound or an image or a video or, you know, plain text, which is what the most common thing here.

You might have to actually pay for some of these things, but they are all out there.

So for example, just a quick example over here, we can take a look at this Open AI Sora package, which is now still in beta mode.

And what it does is it generates videos and you give it a prompt.

For example, the prompt here says the camera directly faces colorful buildings in Burano, Italy.

An adorable Dalmatian looks through a window in a building on the ground floor.

Many people are walking and cycling along the canal streets in front of the building.

And that's the video that is generated by Sora, right?

It's uh oh. It's just the way it is.

Okay, that's how it works. So this will hopefully be available at some point for all of us.

But right now, it's so, uh, being tested out so we can see where Open AI is going.

And I think I should maybe stress over here that almost everything we do is, um, you know, is limited by what ChatGPT or LLMs can do right now.

But the speed with which these things are changing means that you can actually do a lot more with that.

We will be able to do a lot more with it, maybe, you know, just a few months from now.

So keep that in mind. But the basic interaction is going to be fairly, uh, similar to what we are doing today, or the format of the interaction is going to be similar to what we are doing today.

So we said that they follow, uh, like a response model.

You give it an input and it gives you a response. You say something, it gives you a response.

Right. So that's what a chat bot is. And chat bots are applications in general that interface with humans in natural language.

And they communicate in a human human style. That means that it's almost as if you're talking to another human.

They've been around for a while, you know, we have seen them all in our customer service agents on online, which can be really, really frustrating because they never seem to actually answer the question that you have.

But, uh, they have been there for a while. And, uh, but the ones that we have that have been around for a while tend to be very scripted.

That means that they already have a bunch of questions and responses sort of placed inside them.

It understands your question and then gives you a canned response.

You know, that's the general idea. The AI part

Now that we are adding to in chat bots based on LLMs is usually in understanding the user input, and the output is usually canned, right?

That's how these old chatbots are. So Siri and Alexa are examples of chatbots that are voice based.

So you can, you know, see where we are going with this. And um, um, AI as well, but they are, they've been around, so we are used to them.

We already know what they are. Generative AI chat bots generate text based on tokens and the context, and we take a look at this in a second.

They keep track of in-conversation context. That means what they are going to do is they are going to remember what you said before and then use that as a context for answering future questions, right.

So you can set a context and then get a response, and then you can modify the context and it'll sort of take the initial context plus the modification and then respond to you.

Right. That's the idea here. So the key here is this idea of tokens.

And this should be sort of familiar to us. But let's take a quick look at it.

So let's go to this website over here platform.openai.com/tokenizer which is going to show us how ChatGPT is converting our queries into something that it can then use for analysis.

And keep in mind that in the final stage, everything is a number, because that's what computers can work with.

So in this site you're going to see two boxes. And you type in your query or a sentence or whatever in the top box.

So we'll type in Jack Loves Jill over here. And what ChatGPT does is it converts it into tokens.

So each token here corresponds to a word. And that's typically we saw this in our first neural network module as well.

That's typically what is done with text when you're working in any machine learning or AI environment, you convert it into the basic words or the smallest unit of analysis, and you associate numbers with that.

We can see that if you color the tokens we have, Jack is purple, loves is green, Jill is orange.

And in the bottom here it says text and token ID. Then if you see over here this token ID, so click on this, if you click on this you can see that each of these tokens has a number associated with it.

Jack is 33731, loves is 16180, and Jill is 48311.

So that's the basic idea here. This is all familiar to us.

Now the text is converted into tokens. They're just numbers.

And we've converted them into these three numbers over here. Let's flip Jack and Jill and see what happens.

So now we have Jill loves Jack. So Jack no longer or may or may not love Jill, but we know that Jill loves Jack.

So now we notice here that this has got converted into four tokens 41, 484, 16180, and 7762.

So let's take a look at the earlier thing we had here. Jack was 33731, uh, loves was 16180, and Jill was 48311.

Right. So this is the only one, 16180 that is still the same.

Loves is still the same. And Jack and Jill have actually changed their values.

You can think of this as Jack and Jill are now in different contexts.

So in the first um, statement where Jack loved Jill, Jack was the subject and Jill was the object.

And so the context was Jack is the subject who was doing the loving, and Jill is object who is being loved.

And we flipped that around. And now Jill is the subject and Jack is the object, and Jill is doing the loving, and Jack is being loved.

Right. That's the idea here. So because the thought of the interpretation of Jack and Jill has changed, the token values have changed too.

So now we get 41484. And notice that Jill has got divided up into two tokens for some reason.

Right. Which is definitely, you know, possible.

And Jack is now 7762, which was not the original value for Jack.

So this is how ChatGPT is working, right? So you're giving it queries.

Every word in that query is being taken and converted into a token or more than one token.

And uh, that, that the tokens are getting fed into the system.

It's finding some appropriate tokens to return.

And then those appropriate tokens are getting converted back into words and you get a response back.

The context actually matters, right? So we as we can see, because Jack, Jill and Jack have sort of switched around, like I said before.

Right. So Jill and Jack have switched around. The subject object has changed.

And you know, that's the idea here. So tokens are sort of containing or rather are let's say are determined by the context.

So they are not independent of the context in which they are or the position in which they appear in a sentence.

So let's take another example here and let's now say my Aunt Jill loves Jack.

Right. So let's try that over there. So we know that in our if you look at this sentence we can see that my aunt is now the subject.

Right. My Aunt Jill is a subject. Jill and Jack are both objects.

Well Jill is the name of my aunt. And they use tokens in the object context.

And love is to love. So we should see love being the same as before, but everything else being different, I guess.

Kind of. Right. So and we see over here that we have uh, loves is still 16180, which is what it was originally.

My Aunt Jill is now three tokens, my Aunt Jill, which are different from what we thought before just.

Jill and Jack is 7762, which is the same value that we saw when Jack, when Jill loves Jack, was our input because Jack is contextually in the same place.

Even though we have three words over here instead of just one word.

So positionally, Jack is now the fifth word. But in context terms, it's in the same place.

So, you know, it ends up in the same token, if you just do a quick comparison of these three to see that I'm not, you know, lying to you guys.

Sorry is that we get, uh, here, Jack loves Jill, loves was 16180, loves stayed at 16180 and loves stayed at 16180

Because that's really the relationship that's being described between the left hand side and the right hand side, the subject and the object.

And Jack in the subject in the object area stayed the same 7762, 7762 but this kept changing, right?

So this is different. This is different. This is different because each one of these are different.

With that, you can try inserting Jack and Jill in longer sentences and seeing how that changes.

So you know, and give it a try if you will probably find that if the sentence is very long, even if Jack is at the end of the sentence and is the object, it might end up changing meaning because there other objects you know.

Here we have just one subject in this, like either my aunt Jill or just Jill.

But it might end up that, you know, even though Jack is in the same position and has the same general meaning,

it could be if the if you have a very long sentence, then because there are many other objects over there, it might change.

The token might change. So give it a try and play around with it and see a little bit what you get.

As we saw in an earlier module, tokens are the smallest unit of text analysis.

We know that, uh, chatbots generally charge users by the number of tokens.

So this is really why we care about this, right? Because we are going to pay for this stuff.

Finally, if we really want to use, uh, GPT or uh, as AI advances and we start using AI in more complex problems like machine learning or something like that, then you will end up having to pay for it because nothing in this world is for free, unfortunately.

And um, so you have to be cognizant of the fact that every thing that you pay for is based on the number of tokens that you use.

And then the in general, right now, if you're using GPT3, for example, you're limited to 4096 tokens.

If you're using GPT4 it's 8192. GPT4.5 32768.

And each of these is both the prompt as well as the response right.

So combined together, when you ask a question and get an answer, the total number of tokens if you're using, say, GPT4.5 has to be within 32768.

Keep that in mind. So tokens are important because that's the unit of analysis.

And it will be the unit of charging. You know, something like if you were using a cell phone every I guess minute or second counts, I'm not sure it's similarly here it's going to be every token is going to count.

Right. That's the idea there. And that's how it works right now. And then probably in the future as well.

All right. So that's, uh, our background to the, you know, the basics of, uh, LLMs.

And, you know, the basic idea here is that you are going to be working with queries.

These queries are going to be converted into tokens.

You're going to get a response, and you're going to get charged based upon the total size of the query and the response that you get.

Right. So you don't know how long the response is going to be, but, or how many tokens are going to be, but it's going to cost you.